Contrapositive and Contradiction

MTH 317H — Week 5 Notes

1 Contrapositive versus contradiction

An *implication* has the form $P \Rightarrow Q$: if P is true, then Q is true.

The *contrapositive* of $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$.

The two implications $P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ are logically equivalent, meaning they convey the same information.

By contrast, the *negation* of $P \Rightarrow Q$ is $P \wedge \neg Q$. This should not be confused with the contrapositive. Furthermore, $P \implies \neg Q$ and $P \wedge \neg Q$ are also **not** equivalent.

Proof by contrapositive	**Proof by contradiction**
When does it apply? To implications $P \Rightarrow Q$. First identify a hypothesis and a conclusion; rewrite the claim in this form if necessary.	**When does it apply?** To any statement S, including implications and statements not written in if–then form.
What do you assume? Assume $\neg Q$, while keeping all standing assumptions.	**What do you assume?** Assume $\neg S$. If S is $P \Rightarrow Q$, assume *both* P and $\neg Q$, keeping all standing assumptions.
What do you prove? Prove $\neg P$. You are proving the equivalent implication $\neg Q \Rightarrow \neg P$.	**What do you prove?** Derive an impossibility, such as a statement and its negation, or the inequality $1 < 1$.
Must you reach a contradiction? No. A direct proof of $\neg Q \Rightarrow \neg P$ is enough.	**Must you reach a contradiction?** Yes. State precisely which facts are incompatible.
How do you finish? "This proves the contrapositive, so the original implication follows."	**How do you finish?** "Thus $\neg S$ is impossible, so S is true." For an implication, conclude that P cannot hold with $\neg Q$.
When is it useful? When $\neg Q$ gives a convenient starting point and $\neg P$ is easier to establish than Q.	**When is it useful?** When denying the claim gives assumptions that cannot all be true, including for nonexistence or irrationality claims.

Do not confuse the contrapositive with the converse. The converse is $Q \Rightarrow P$, which need not be equivalent to $P \Rightarrow Q$.

2 Writing a proof by contrapositive

Procedure. Identify P and Q, state the contrapositive, assume $\neg Q$, and prove $\neg P$. Keep the domain and other standing hypotheses. For integers, "not odd" means even and "not even" means odd.

Example 1. The same implication proved in two ways

Let $x \in \mathbb{Z}$. Prove that if $5x - 11$ is even, then x is odd. Here P is "$5x - 11$ is even" and Q is "x is odd."

Proof by contrapositive. We prove that if x is even, then $5x - 11$ is odd. Assume that x is even. Then $x = 2k$ for some $k \in \mathbb{Z}$, so

$$5x - 11 = 10k - 11 = 2(5k - 6) + 1.$$

Since $5k - 6 \in \mathbb{Z}$, the number $5x - 11$ is odd. This proves the contrapositive, hence the original implication. □

Proof by contradiction. Suppose that $5x - 11$ is even and x is not odd. Since $x \in \mathbb{Z}$, it is even, so $x = 2k$ for some $k \in \mathbb{Z}$. Then

$$5x - 11 = 2(5k - 6) + 1,$$

which is odd. This contradicts the assumption that $5x - 11$ is even, because no integer is both even and odd. Therefore x is odd. □

Compare the assumptions. The first proof assumes only that x is even, in addition to $x \in \mathbb{Z}$. The second also assumes that $5x - 11$ is even. Both use the same calculation, but their conclusions have different roles.

Example 2. Negating an "or" statement

Let $x, y \in \mathbb{R}$ with $x, y \geq 0$. Prove that if $xy > 100$, then $x > 10$ or $y > 10$.

Proof. We prove the contrapositive. The negation of "$x > 10$ or $y > 10$" is "$x \leq 10$ and $y \leq 10$." Assume $x \leq 10$ and $y \leq 10$. Keeping the standing assumptions $x, y \geq 0$, we obtain

$$xy \leq 10y \leq 100.$$

Thus xy is not greater than 100, proving the contrapositive. □

Example 3. Contradiction for a statement not written as an implication

Claim. There is no largest integer.

Proof. Suppose there is a largest integer M. Then $M+1$ is an integer and $M+1 > M$, contradicting the choice of M. Hence there is no largest integer. □

3 Common mistakes and practice

Check the structure of your proof

- **Negate the whole conclusion.** The negation of $A \vee B$ is $\neg A \wedge \neg B$; the negation of $A \wedge B$ is $\neg A \vee \neg B$. Also, the negation of $x > c$ is $x \leq c$.
- **Keep the domain.** In a claim about integers, the contrapositive still concerns integers. Do not negate a standing hypothesis when reversing the implication within that domain.
- **Negate quantifiers correctly.** To contradict $\forall n \in \mathbb{N},\ P(n)$, assume that there is some $n \in \mathbb{N}$ for which $P(n)$ fails, not that it fails for every n.

Practice problems

For each proof, state your method and finish with a sentence explaining why the original claim follows.

1. Let $x \in \mathbb{Z}$. Write the contrapositive of the following implication and prove it:

$$x^2 - 6x + 5 \text{ is even} \quad \Longrightarrow \quad x \text{ is odd.}$$

2. Let $x > 0$. Prove that if $x - 2/x > 1$, then $x > 2$. Give one proof by contrapositive and one by contradiction. State the assumptions in each proof. Be sure to include $x = 2$ when you negate the conclusion.

Hints

1. Substitute $x = 2k$. **2.** If $0 < x \leq 2$, then $2/x \geq 1$.